UNIVERSITY OF GOROKA

MCS444-Windows server 2022 Lab

Title:Installing and configuring windows server in Oracle VM VirtualBox

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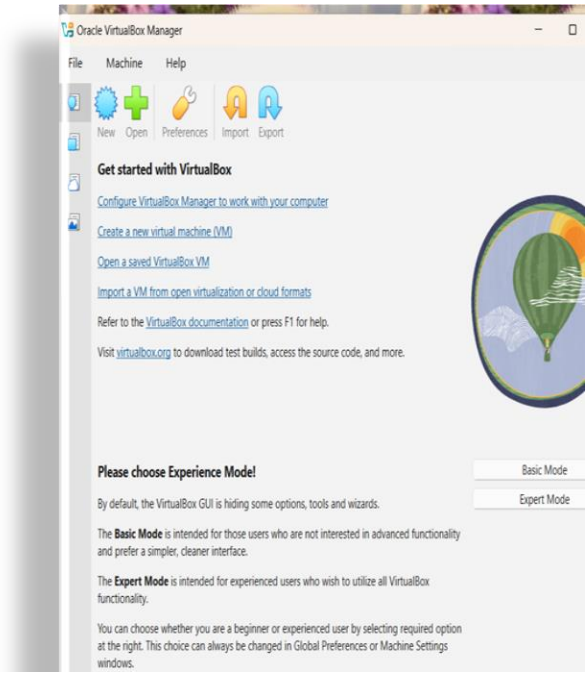


This report documents the practical lab for installing and configuring windows server 2022 in oracle VM virtual box, the lab required creating a virtual machine, attaching it to a NAT network named MCS Nat, configuring a static IPV4 address, and using windows command line to verify connectivity. The purpose was to practice virtualization, server deployment, and ipv4 networking. Below are the steps with screenshot attached.

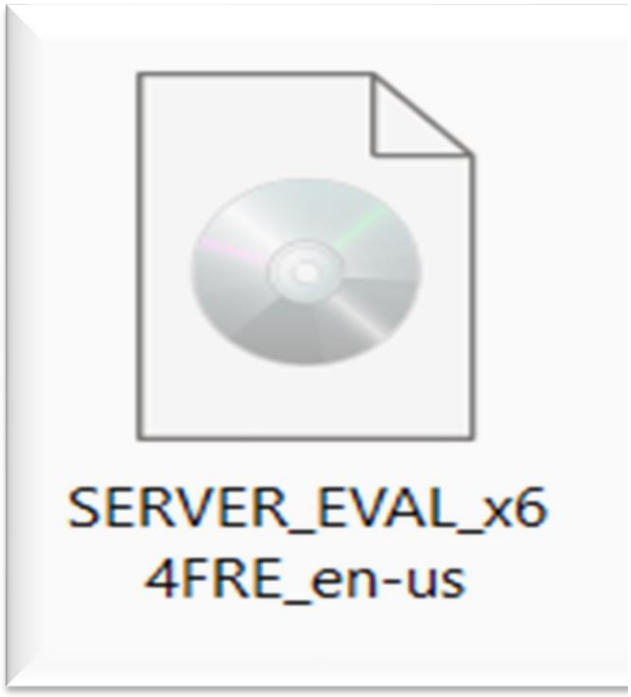
Before doing any practical lab, there must be a lab set up, so firstly I checked that my computer has free ram and disk space to run a virtual machine, and then I went to the official website (https://www.virtualbox.org/), downloaded and installed oracle VM virtual Box, after installing oracle VM, I went to https://www.microsoft.com/evalcenter/ and downloaded the Windows Server 2022 ISO file

Below is the screen shot of the installed oracle VM virtual box and the downloaded ISO file.

Installed Oracle VM

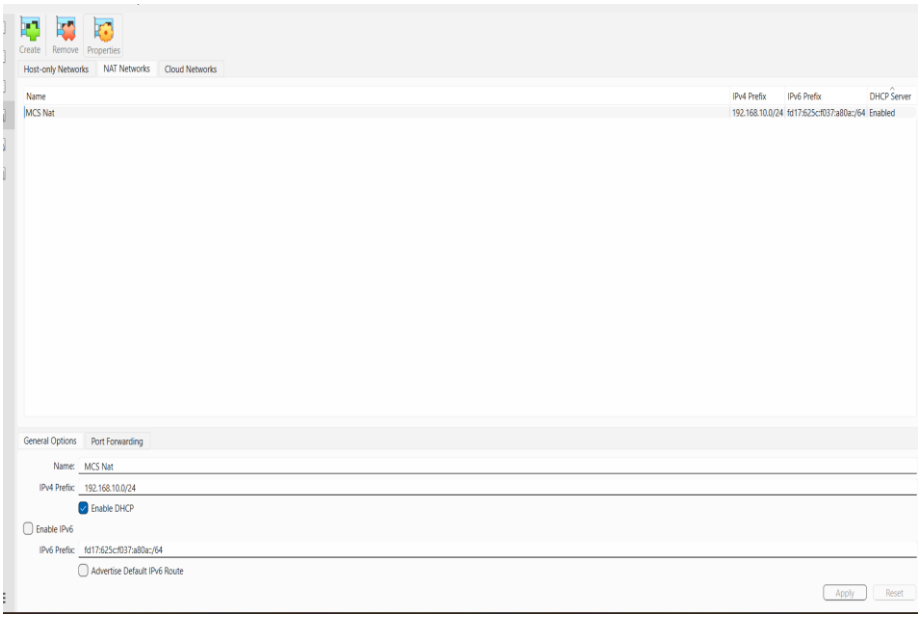


Downloaded window server ISO file



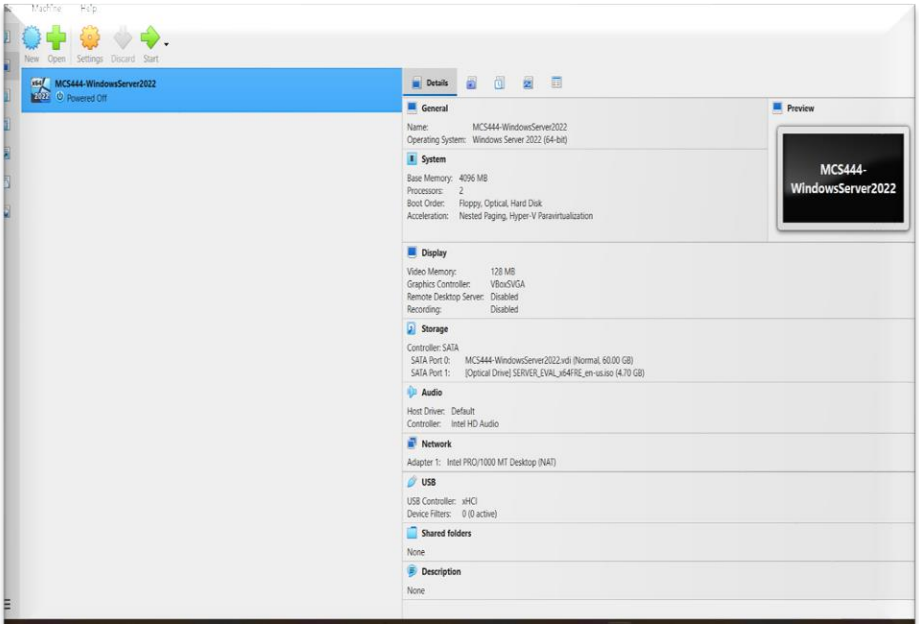
After the installation and download of the windows server, I opened the virtual box and clicked on the network setting which is the 5th image on the left column, I selected the NAT network tab, chose create and named the network MCS Nat, and allocated the IPV4 address 192,168.10.0/24 and saved it or clicked apply, below is the screenshot of the creation of the MCS Nat Network.

MCS Nat NETWORK CREATED



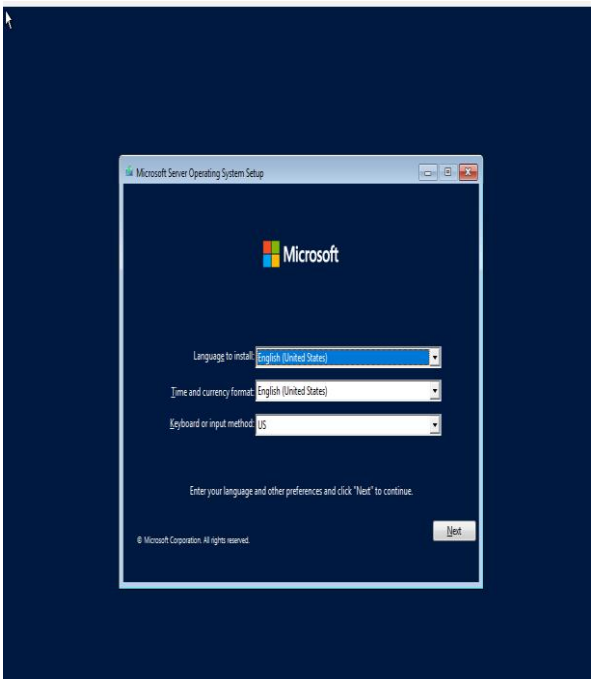
Then I Clicked machine, selected create and created a virtual machine named MCS444-WindowsServer2022, under installation image, I selected the windows server 2022 ISO file, then I chose Microsoft windows as the type and windows 2022 (64 bit) as the version, I assigned 2 virtual CPUS,4 GB ram, used 8 GB for memory and 60 GB for storage, below is the screenshot of what I've just created.

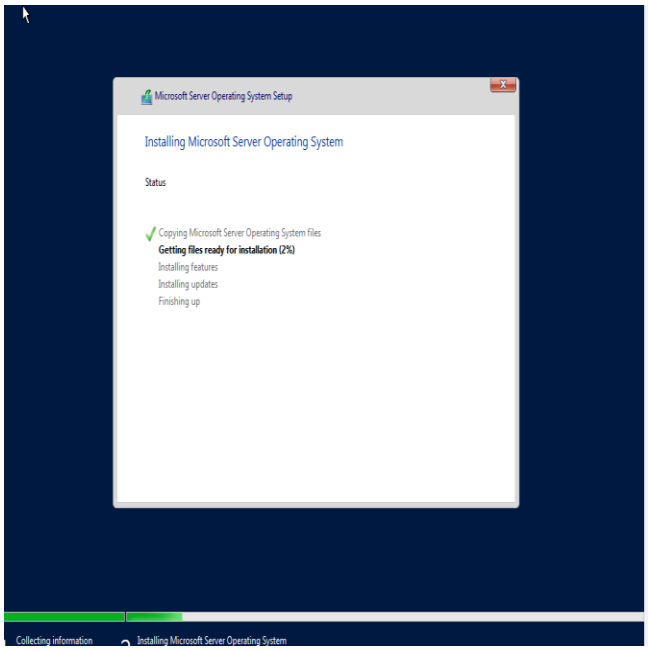
NEW VM CONFIGURATION



After the creation of a new virtual machine, I clicked start and selected the windows server 2022 standard evaluation (Desktop Experience), which I forgot to take a screen shot but, I followed the installation prompts and waited for the installation to complete, below is the screen shot of the first prompt screen and the installation process of Microsoft server OS set up.

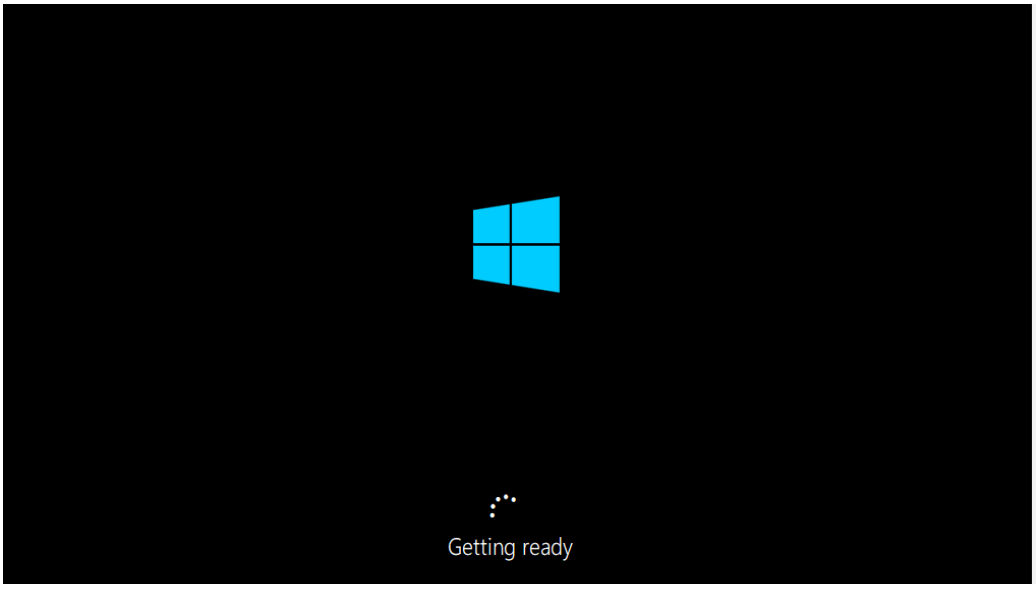
WINDOWS SERVER SETUP





Below is the screen shot of the first boot.

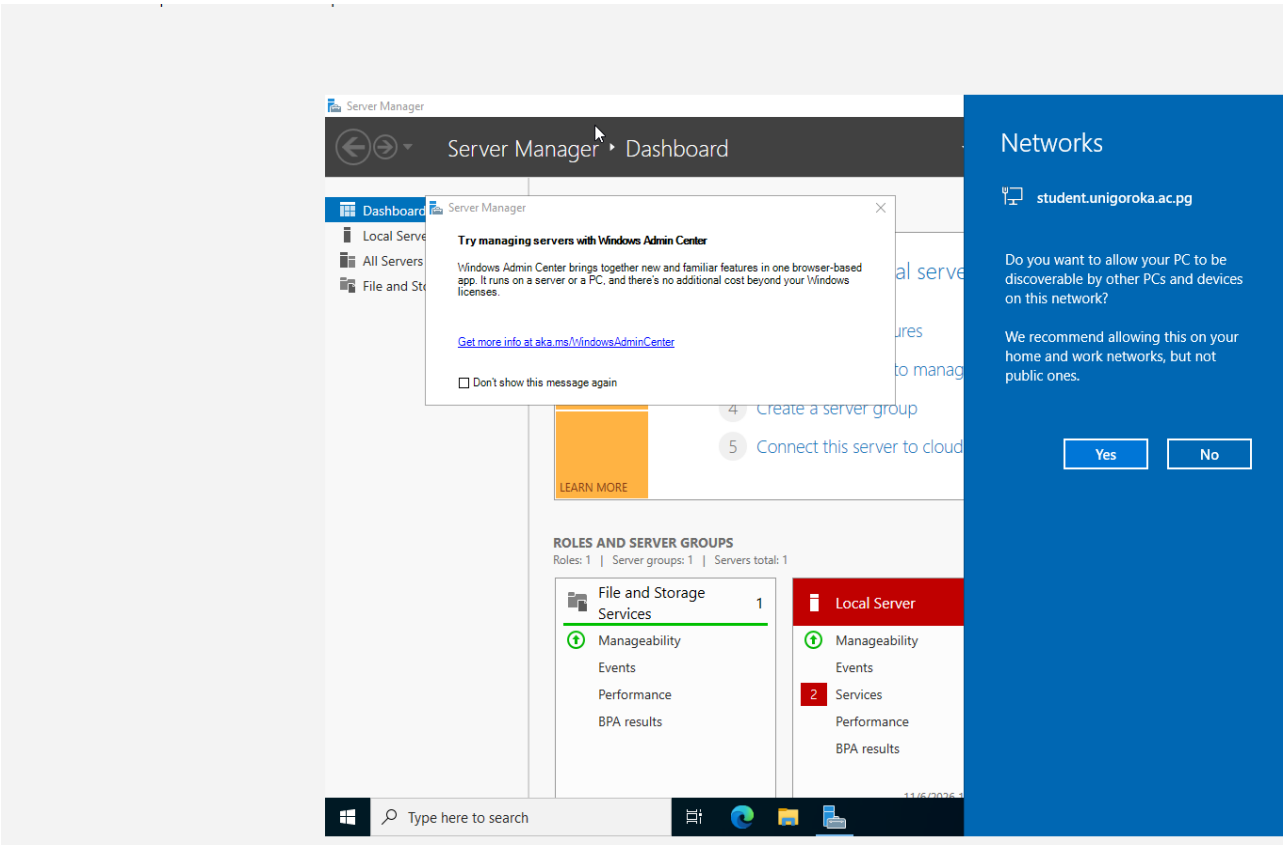
FIRST BOOT





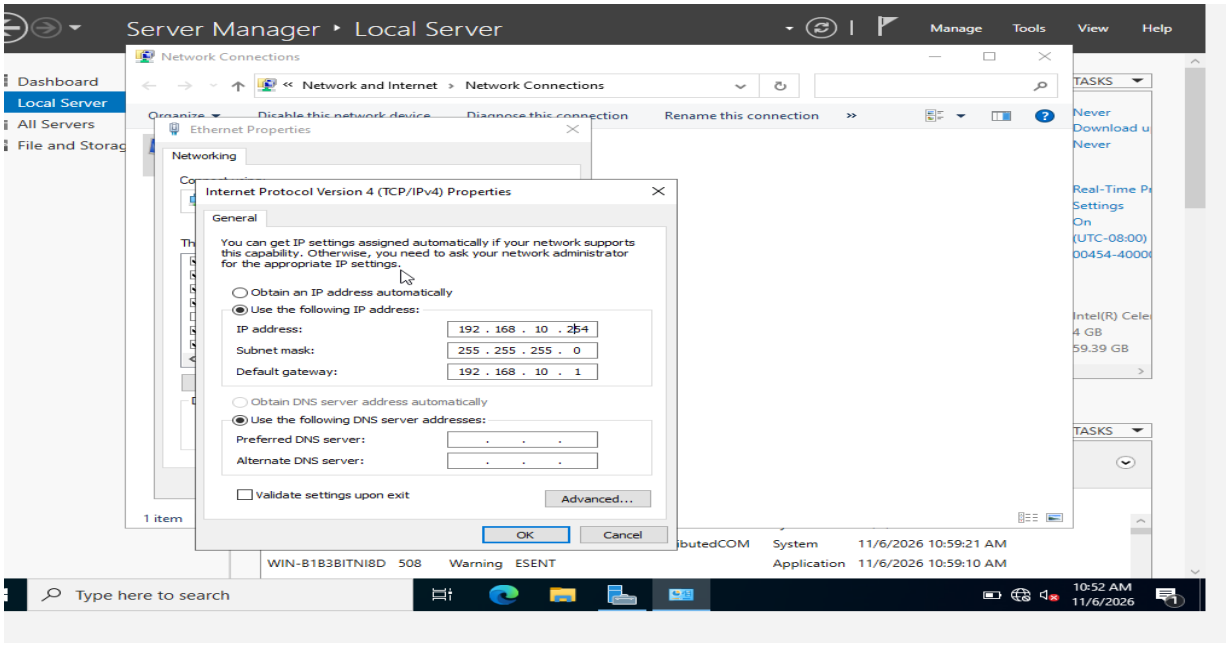
Then I set the administration password and logged in to the server desktop, below is the screen shot of the windows server desktop after successful installation.

WINDOWS SERVER DESKTOP



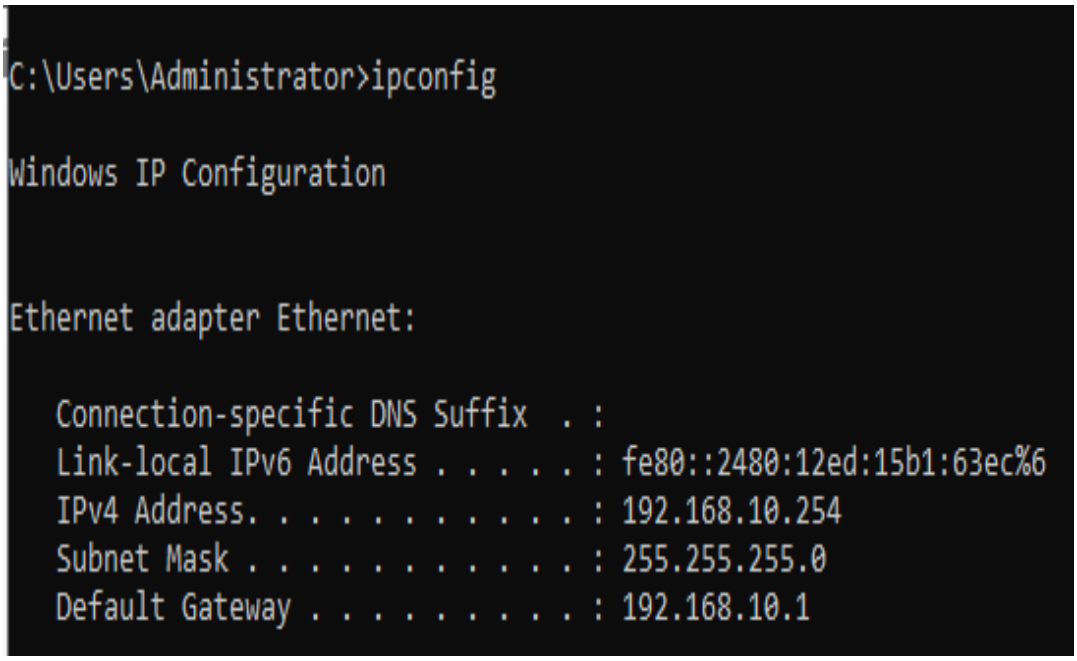
After logging in, I clicked on the local server, opened network connections, right-clicked ethernet, and I opened properties. I changed the setting from obtaining Ip address automatically to use the following address and entered 192.168.10.254, the subnet mask automatically appeared and I set the default gate way to 192.168.10.1, then I clicked ok and closed the setting. Below is the screen shot of IPV4 properties showing static IP configuration 192.168.10.254.

STATIC IPV4 PROPERTIES



To verify the ipv4 configuration that I did, I went to my search bar and typed CMD and opened command prompt, I ran ipconfig command, the result showed the ethernet adapter was configured with ipv4 address 192.168.10.254, subnet mask 255.255.255.0 and the default gateway 192.168.10.1, this confirmed that the static IP configuration was applied successfully. Next, I typed the command ping 192.168.101, to test the connectivity to the gateway, the result showed o% loss, which confirmed that the server could successfully communicate with the gateway on the MCS Nat network. Below are the screen shots of ipconfig result and the ping results.

IP CONFIG/ ALL OUTPUT



GATEWAY CONNECTIVITY TEST

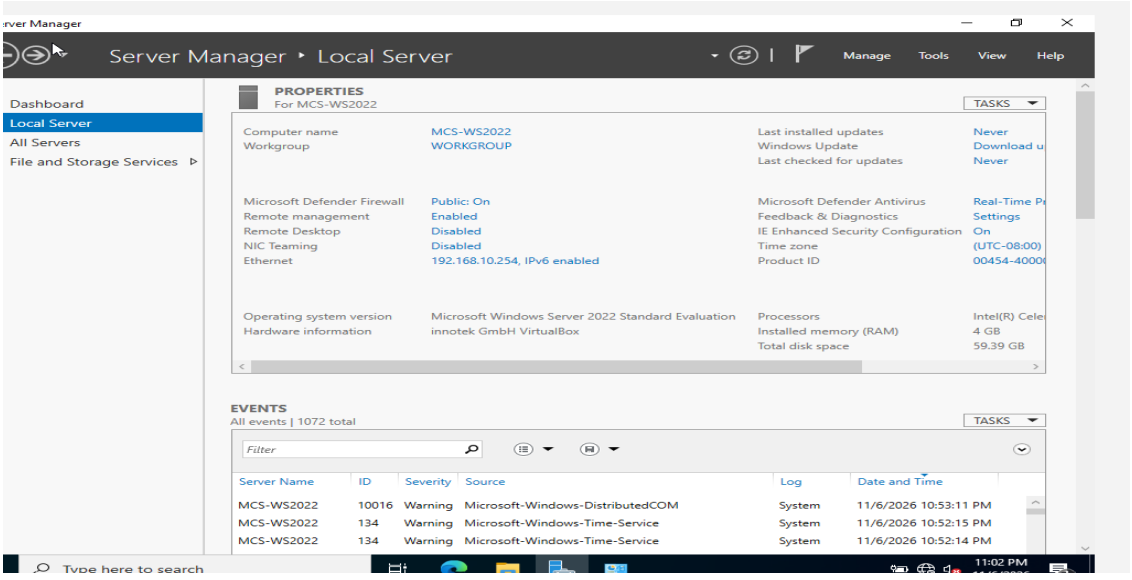
```
C:\Users\Administrator>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128

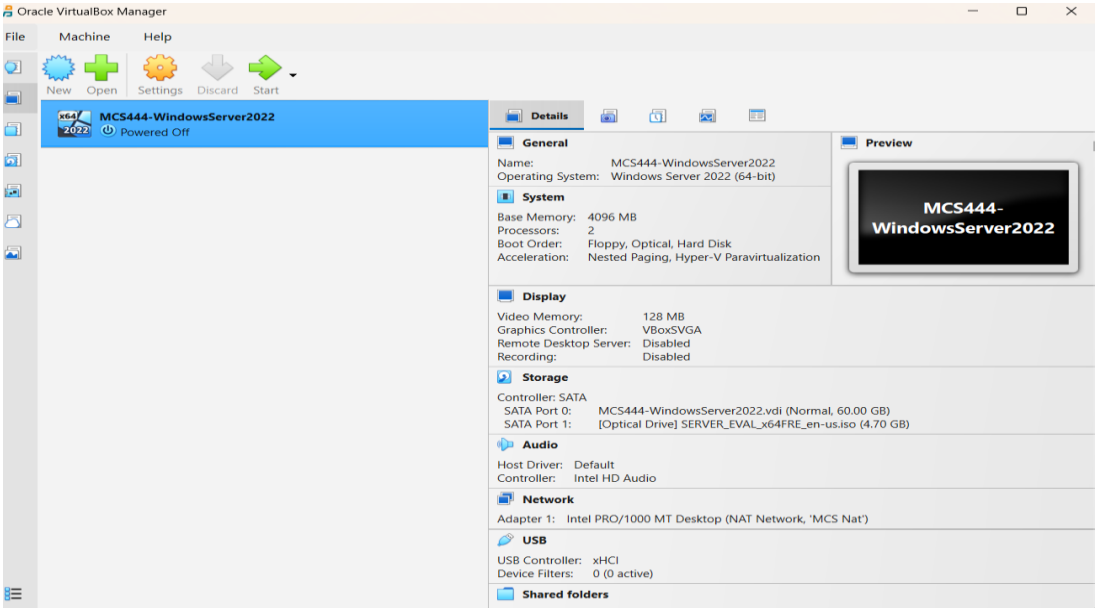
Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

After verifying the configuration, I exited the command prompt, went to the local server, clicked the current computer's name and renamed it as MCS-WS2022 and clicked confirmed and restarted my machine when prompted, after restarting it I signed in and confirmed that my computers named has been renamed to MCS-WS2022 under system properties or server manager, below is the screen shot of the renamed computer name.

COMPUTER NAME



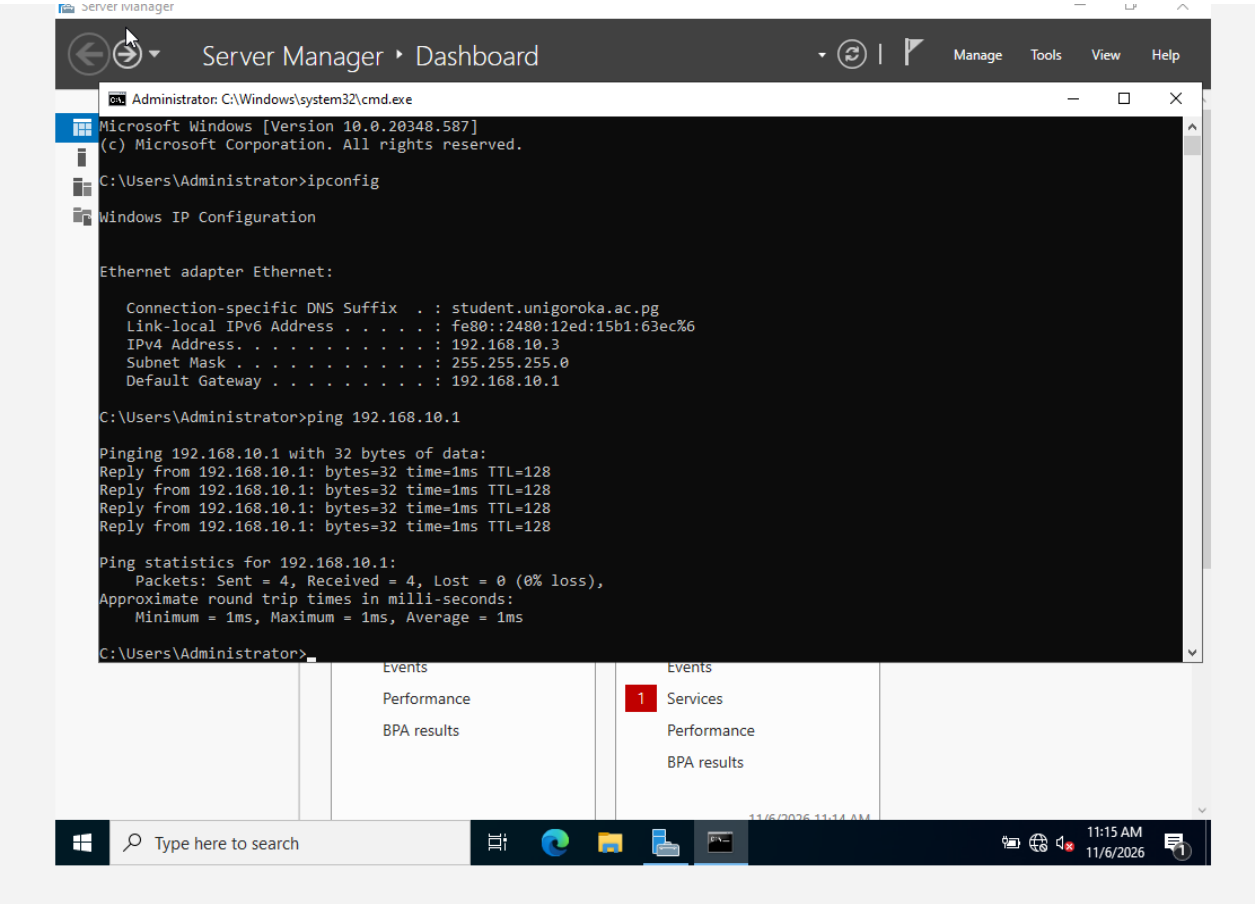
Finally, I went to the start menu and shut down my virtual machine, below is the screen shot of the installed VM In virtual box.



TROUBLE SHOOTING

Initially I miss entered the IP address 192.168.10.254 as 192.168.10.3 and saved it, it saved successfully because the gate way was 192.168.10.1, I didn't realize until I entered ipconfig command on command prompt, the result showed the ethernet adapter was configured with ipv4 address 192.168.10.3, subnet mask 255.255.255.0 and the default gateway 192.168.10.1 and still showed successful connection when ping, because the IP and the gate way IP start with same numbers even though, the IP address assigned is not on the lab practical, however using a low IP address for a server is not ideal. If many PCs and servers are all given low addresses, the low range will quickly run out and DHCP may assign the same addresses to another devices, causing an IP conflict.

To fix it, I exited command prompt, went to the network connection, Ethernet, right clicked and entered properties, I opened internet protocol version 34 (TCP/IPV4) and change the IP address from 192.162.10.3 to 192.168.10.254, as stated in the lab work and clicked OK, then I exited and opened command prompt to check if the IP address was changed successful and has a successful connection, so after entering ipconfig and ping 192.168.10.254 ,the IP address was changed and connection was successful, below is the screenshot of the successful connection using the wrong IP address.



Through this lab, I have gained insights on virtualization where I used virtual box to instal and run windows server2022 inside my laptop, static IP configuration where, I manually assigned the ipv4 address that will not change , gateway verification using command prompt to check for successful connection and trouble shooting.